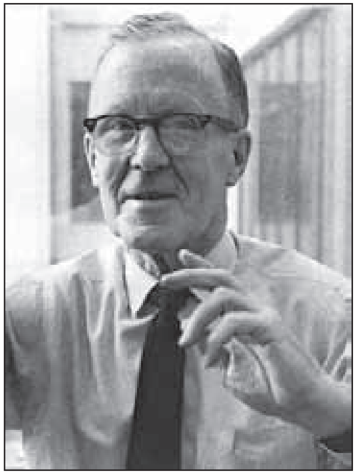
systems has hence come to rely on the human brain as a yardstick. As we saw with the Turing Test, an intelligent agent is deemed so only if it manages to fool a human into thinking it's human.

With that goal always at the back of their minds, scientists have sought to emulate the human mind in machines, albeit with different paths and results. The earliest form of these efforts meant studying computational neuroscience – the science of how the human brain and nervous system processes information – and applying it to electronics.



Donald Hebb

A lot of the initial work was based on the findings of Donald Hebb, a neuropsychologist at the McGill University in Canada. In the 1930s, he was one of the first to talk about "learning" from a neural perspective. Hebb said learning was all about the neural networks i.e. the clusters of cells and neurons. The strength of the connections between these cells and neurons affected how a human brain learns. To this end, he theorised a mechanism by which the brain learns to distinguish objects and signals, add and understand grammar.

Hebb's followers took his ideas and reshaped them into two broad concepts:

i) **Autonomous systems:** Each neuron is a self-adjusting cell that changes the strengths of its connections to other neurons when learning. This way, fewer errors are made when it repeats a task.

ii) **Instantaneous learning:** Scientists also presumed that learning in the brain is "instantaneous", i.e. as soon as something to be learned is presented, the appropriate brain cells use their "local learning laws" to make instant adjustments to the strength of their connections to other neurons. When learning is complete, the brain discards its memory of the learning example.

Neurology became the focus of researchers in the artificial intelligence fields and soon, they started building machines based on the wiring of